

Customer Journey Map

5. PROJECT DEVELOPMENT PHASE

Date	4 July 2026
Team ID	XXXXXX
Project Name	A Comprehensive Measure of Well Being
Maximum Marks	3 Marks

Code-Layout, Readability, and Reusability Report

This document reviews our codebase structure, coding conventions, readability metrics, and modular design.

1. Code-Layout Conventions

We adhere to the **PEP 8 style guide** for Python development: * **Naming Conventions:** * Variables and functions: lowercase with underscores (`life_expectancy`, `load_ml_resources`). * Class names: PascalCase (`Config`). * Constants: uppercase with underscores (`BASE_DIR`, `RAW_DATA_URL`). * **Indentation:** 4 spaces (no tabs). * **Maximum Line Length:** Capped at 80-120 characters for clean terminal viewports. * **Blank Lines:** Two blank lines surrounding class functions and module-level code.

2. Readability & Code Documentation

- **Docstrings:** All python functions include detailed multi-line docstrings detailing the purpose, arguments, and return types.

- **Logging:** Avoid using raw `print()` statements for production logging. Built-in logging module is configured in `app.py`, writing formatted event traces to standard stdout and a persistent `app.log` file.
 - **Self-documenting variables:** Variables are explicitly named (e.g. `expected_years_schooling` instead of `ey`s), enhancing code transparency.
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3. Reusability Design Patterns

1. Jinja2 Templates Inheritance:

2. [base.html](#) serves as the parent layout structure containing the dark mode toggle scripts, mobile navigation responsive drawer, loading screen, and headers.
3. Target views ([index.html](#), [result.html](#), [about.html](#)) extend `base.html`, overriding only their specific content blocks.

4. Unified Configuration Class:

5. [config.py](#) isolates folder directories, dataset links, and serialization paths in a single class imported across script pipelines.

6. Reusable CSS Classes:

7. Custom helper variables (`--teal-primary`, `--bg-app`) are defined in `:root`. Modern design tokens like glassmorphism card wrappers (`.glass-card`, `.glass-panel`) are written once and used globally across templates.